

[Manganese: Fact Sheet for Consumers \(original version\)](#)

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What is manganese and what does it do?

Manganese is a mineral that your body needs to stay healthy. Your body uses manganese to make energy and protect your cells from damage. Your body also needs manganese for strong bones, reproduction, blood clotting, and a healthy immune system.

How much manganese do I need?

The amount of manganese you need depends on your age and sex. Average daily recommended amounts are listed below in milligrams (mg).

Life Stage	Recommended Amount
Birth to 6 months	0.003 mg
Infants 7–12 months	0.6 mg
Children 1–3 years	1.2 mg
Children 4–8 years	1.5 mg
Boys 9–13 years	1.9 mg
Girls 9–13 years	1.6 mg
Teen boys 14–18 years	2.2 mg
Teen girls 14–18 years	1.6 mg
Adult men	2.3 mg
Adult women	1.8 mg
Pregnant teens and women	2.0 mg
Breastfeeding teens and women	2.6 mg

What foods provide manganese?

Many foods contain manganese. You can get recommended amounts of manganese by eating a variety of foods, including the following:

- Whole grains, such as brown rice, oatmeal, and whole-wheat bread
- Clams, oysters, and mussels
- Nuts, such as hazelnuts and pecans
- Legumes, such as soybeans and lentils
- Leafy vegetables, such as spinach and kale
- Some fruits, such as pineapple and blueberries
- Tea
- Many spices, such as black pepper

What kinds of manganese dietary supplements are available?

Manganese is available in many multivitamin/mineral and other dietary supplements. Supplements contain many forms of manganese, such as manganese sulfate and manganese aspartate. Scientists don't know whether any form of manganese in supplements is better than any other form.

Am I getting enough manganese?

Most people in the United States get enough manganese from the foods they eat.

What happens if I don't get enough manganese?

Manganese deficiency is very rare in the United States. A deficiency might cause the following symptoms:

- Weak bones and poor growth in children
- Skin rashes and loss of hair color in men
- Mood changes and worse premenstrual pain than normal in women

What are some effects of manganese on health?

Scientists are studying manganese to understand how it affects health. Here are some examples of what this research has shown.

Bone health

You need manganese in combination with other minerals and vitamins for healthy bone formation. However, more research is needed to understand the role of manganese in maintaining or improving bone health.

Diabetes

You need manganese to help break down the starches and sugars that you eat, but its effect on the risk of diabetes is unknown. More research is needed to understand whether manganese plays a role in the development of diabetes.

Can manganese be harmful?

Studies have not shown any harm from the manganese in food and beverages, but some people have developed manganese toxicity by consuming water containing very high levels of manganese. Another cause of manganese toxicity is inhaling large amounts of manganese dust from welding or mining work.

The symptoms of manganese toxicity include tremors, muscle spasms, hearing problems, mania, insomnia, depression, loss of appetite, headaches, irritability, weakness, and mood changes.

People should not consume more manganese than the upper limits from food, beverages, or dietary supplements unless their health care provider recommends doing this.

The daily upper limits for manganese include intakes from all sources—food, beverages, and supplements—and are listed below.

Ages	Upper Limit
Birth to 6 months	Not established
Infants 7–12 months	Not established
Children 1–3 years	2 mg
Children 4–8 years	3 mg
Children 9–13 years	6 mg
Teens 14–18 years	9 mg
Adults	11 mg
Pregnant and breastfeeding teens	9 mg
Pregnant and breastfeeding women	11 mg

Does manganese interact with medications or other dietary supplements?

Manganese is not known to interact or interfere with any medicines.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if the dietary supplements might interact with your medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients, such as manganese.

Manganese and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about manganese?

- For more information about manganese
 - Office of Dietary Supplements (ODS) [Health Professional Fact Sheet on Manganese](#)
- For more information on food sources of manganese
 - USDA, [FoodData Central](#)
- For advice on choosing dietary supplements
 - ODS, [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

cell

The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

consume

To eat or drink.

deficiency

An amount that is not enough; a shortage.

depression

A serious medical condition that can interfere with an individual's ability to work, study, sleep, and eat. Symptoms include ongoing feelings of sadness and despair, loss of energy, and loss of interest or pleasure in activities that were once enjoyed. A person who has depression may experience changes in eating or sleeping habits, and have thoughts of death or suicide.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

immune system

A group of organs and cells that defends the body against infection, disease, and altered (mutated) cells. It includes the thymus, spleen, lymphatic system (lymph nodes and lymph vessels), bone marrow, tonsils, and white blood cells.

infant

A child younger than 12 months old.

insomnia

Difficulty in going to sleep or in getting enough sleep.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

legume

Dried beans and peas, including kidney beans, pinto beans, black beans, navy beans, lima beans, black-eyed peas, garbanzo beans (chickpeas), split peas, and lentils. Legumes are good sources of protein, iron, zinc, dietary fiber, folate, and potassium.

milligram

mg. A measure of weight. It is a metric unit of mass equal to 0.001 gram (it weighs 28,000 times less than an ounce).

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

Multivitamin/mineral dietary supplement

MVM. A product that is meant to supplement the diet. MVMs contain a variety of vitamins and minerals. The number and amounts of these nutrients can vary substantially by product.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by

evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

prescription

A written order from a health care provider for medicine, therapy, or tests.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

soy

A plant that produces beans used in many food products. Soy products contain isoflavones (estrogen-like substances) that are being studied in the prevention of cancer, hot flashes that occur with menopause, and osteoporosis (loss of bone density). Also called soya and soybean. Latin name: *Glycine max*.

spasm

An abnormal and uncontrollable muscle contraction or cramp.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

tea

A drink made by adding boiling water to fresh or dried herbs and steeping (soaking) them. It may be drunk either hot or cold. Also called an infusion.

toxicity

The degree to which something is poisonous (toxic).

tremor

A trembling or shaking in one or more parts of the body, usually the hands. An individual can also have tremors in the arms, head, face, vocal cords, torso, and legs.

upper limit

UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

whole grain

Unprocessed seeds of edible grasses, including brown rice, buckwheat, bulgur, millet, popcorn, oats, quinoa, whole-grain barley, whole rye, whole wheat, and wild rice. Grains that are ground, cracked, or flaked can be labeled whole grain if they have the same amount of bran, germ, and endosperm (the inner part of the seed kernel) as the intact grain. Whole grains are sources of iron, magnesium, selenium, B vitamins, and dietary fiber. Eating whole grains may help lower the risk of heart disease, obesity, and type 2 diabetes.

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